

EDUCATION

- Rutgers University

PhD degree in Bioinformatics

Conduct RNA-seq analysis to explore gene expression profiles in cancer patients (TCGA, dbGaP, GEO, cBioportal)

Develop computational methods to identify biomarkers for cancer therapeutics

Apply systems biology methods to integrate transcriptomics data for mechanistic insights into cancer progression.

Newark, NJ, US

2022 – Present
- Moscow Institute of Physics and Technology, MIPT

Master’s degree in Biophysics and Bioinformatics

Moscow, Russia

2019 – 2021
- Moscow Institute of Physics and Technology, MIPT

Bachelor’s degree in Applied Math and Physics

Moscow, Russia

2015 – 2019

RESEARCH AND WORK EXPERIENCE

- Alma AI, Inc. (startup)

Intern in machine learning and computational biology

Python

Protein DL models

Fine-tuning protein models by preparing datasets, running training pipelines, and benchmarking model performance.

Analyze results, identify failure cases, and propose improvements to enhance predictive accuracy on therapeutic tasks.

NYC, US

Present
- Johnson and Johnson

Immunology Translational Research Scientist Intern

Single cell analysis

R

Co-expression analysis

Conducted expression and co-expression analysis of inhibitory receptors across immunological diseases.

Analyzed single cell datasets from publicly available sources.

Spring House, PA, US

summer 2025
- Ayil Bank

Data Scientist

Python

SQL

PowerBI

Excel

Data Analytics

Crafted and implemented robust pipelines for streamlined data collection from bank databases using Python and SQL, ensuring reliable and efficient data integration.

Developed PowerBI dashboards and reports for real-time data monitoring, significantly reducing data generation time to an hour and enhancing operational efficiency.

Bishkek, Kyrgyzstan

2022
- Skolkovo Institute of Science and Technology, Skoltech

Research Fellow. Project: Graph-based assessment of the structural significance of aminoacid residues

Python

R

bash

Gromacs

PyMOL

Structural Bioinformatics

Moscow, Russia

2019 – 2021
- Institute of Science and Technology Austria, ISTA

Project: Particle picking optimization in cryo-EM images, Research intern

Python

bash

TOPAZ

crYOLO

Vienna, Austria

summer 2019

PUBLICATIONS

- Machine learning framework for mRNA alternative splicing analysis identifies a signature of progression in colorectal adenocarcinoma (in review)

U.Maimekov

M.Nosrati

A.Mahmoud

M.Mustafi

M.Craige

F.Coffman

J.S.Parrott

C.Lutz

A.Mitrofanova

2025

Reducing the concentration of Benzo(a)pyrene in the soot particles in gas phase by using and burning water fuel emulsions

K.Kemelov

U.Maimekov

D.Sambaeva

Z.Maymekov

2020
- CONFERENCES AND COMPETITIONS
- The 2025 Annual Retreat on Cancer Research in New Jersey

Gallo Award for outstanding cancer research

May 22, 2025

NYC Biohack: Developed GPT-based model for gene info extraction

Oct. 2024

RECOMB Conference, Boston, MA

Poster: Mechanism-centric network-based machine learning algorithm for predictive modeling in oncology

May 2024

Rutgers Hackathon: AI Ureteral Injury Prevention System

Oct. 2023

Silver medal on Kaggle: NeurIPS 2024 - Predict New Medicines with BELKA

July 2024

Bronze medal on Kaggle: CAFA 5 Protein Function Prediction

Dec. 2023

- **Online webinar: Data Science application in Bioinformatics**
Invited by **The Cramer Project** company, Bishkek, Kyrgyzstan *May, 2023*
- **13th Annual SHP Student Research and Scholarship Symposium** , Newark, NJ
"Alternative Splicing Events Predict Overall Survival in Colorectal Adenocarcinoma" *May, 2023*

EXTRACURRICULAR EXPERIENCE

- E-board member, Data Science Club, Rutgers University, RBHS *2023 – Present*
 - Coordinated and organized data science workshops, guest lectures,
 - Developed educational content, including tutorials and presentations on Python, R, and data analysis techniques